

Shweta U. Narkar, Ph.D.

+1 (646) 239-5325 | shweta.3695@gmail.com | linkedin.com/in/shwetanarkar | narkarshweta.com | [Google Scholar](https://scholar.google.com/citations?user=shweta)

RESEARCH INTERESTS

AI/ML approaches for complex scientific and environmental systems; integration and curation of heterogeneous geochemical, environmental, and experimental datasets; ontology engineering and knowledge graphs as structured scaffolding for scientific ML; explainable and interpretable machine learning for data-intensive Earth and environmental research; reproducible computational workflows and FAIR data practices for community-driven scientific infrastructure.

EDUCATION

Rensselaer Polytechnic Institute

Troy, NY

Ph.D., Computer Science (Multidisciplinary: Computer Science & Astrobiology)

May 2025

- Advisors: Dr. Peter A. Fox (-2021), Dr. James A. Hendler and Dr. Karyn L. Rogers

Rensselaer Polytechnic Institute

Troy, NY

M.S., Data Science & Analytics

May 2019

- Advisor: Dr. Peter A. Fox

University of Pune

Pune, India

B.E., Computer Engineering

Aug 2017

TECHNICAL SKILLS

Programming: Python, R

ML & Data Science: scikit-learn, pandas, NumPy, PyTorch, SHAP, statistical modeling, model evaluation

Semantic Web & Ontologies: OWL/RDF, RML Mapping, SPARQL, Protégé, HermiT, Pellet, GraphDB, RMLMapper

Data Systems & Tools: MongoDB, PostgreSQL, Git, Docker, Azure DevOps, Jupyter, GCP, GitHub, Express.js

Visualization: Matplotlib, Seaborn, ggplot2, D3.js

Other: Hugging Face Transformers, LLMs, FAIR data principles

POSITIONS HELD

Postdoctoral Research Associate

Aug 2025 – May 2026

Rensselaer Astrobiology Research and Education (RARE) Center, RPI

Troy, NY

- Prepared manuscript on POLO (Prebiotic Origins of Life Ontology) for submission to the International Semantic Web Conference (ISWC 2026).
- Collaborated with experimental teams to transform laboratory workflows into structured, ML-ready datasets for downstream modeling and analysis.
- Designed and curated scientific datasets enabling reproducible data pipelines and predictive modeling.
- Mentored 6 researchers across undergraduate, graduate, and high school levels in research methodology, data analysis, and scientific communication.

AI/ML & Data Scientist

Feb 2026 – Present

Mirro.ai

Remote

- Developing a benchmarking pipeline in Python to evaluate speech emotion recognition model performance against labeled audio datasets (IEMOCAP, MSP-Podcast).
- Benchmarked MedGemma 4B (4-bit quantization, NF4/bfloat16) on SOAP note generation; achieved ROUGE-1: 0.6053, ROUGE-2: 0.3405, ROUGE-L: 0.4107.
- Identified dataset-model co-development bias in EmoNet-Voice Bench (ICLR 2026 withdrawal); advocated for IEMOCAP and MSP-Podcast as community standard alternatives.

PhD Research Assistant, Applied ML & Data Science

May 2021 – May 2025

Rensselaer Polytechnic Institute - NASA-funded Earth First Origins project

Troy, NY

- Developed POLO (Prebiotic Origins of Life Ontology), the first formal ontology for experimental prebiotic chemistry: 19 classes, 7 object properties, 41 data properties, 12,000+ RDF triples via declarative RML mapping pipeline, enabling structured annotation and reuse of 200+ hydrothermal experiments.
- Trained ML models (Random Forest, Gradient Boosting) on curated prebiotic chemistry data, achieving $R^2 = 0.734$ and informing the design of 24 follow-up experiments.

- Built EORE (Executable Online Resource of Experimental data), a reusable data ingestion framework (Express.js, MongoDB) capturing 190+ experiments for downstream modeling.
- Collaborated with experimental scientists to analyze datasets and develop visualizations supporting hypothesis generation.

Machine Learning Research Intern

May 2020 – Sept 2020

IBM Research

Yorktown Heights, NY

- Designed and implemented Model LineUpper, a system for multi-level comparison of ML models enabling expert evaluation beyond aggregate performance metrics.
- Led a controlled user study with 14 professional data scientists evaluating 16 ML models; translated findings into actionable insights for technical and non-technical stakeholders.
- Applied SHAP-based explainable AI techniques to analyze model behavior and support instance-level error diagnosis.
- Co-authored peer-reviewed research published at ACM IUI 2021 (42 citations).

PhD Research Assistant, Scientific Data Systems

Jun 2018 – Apr 2021

Tetherless World Constellation, Rensselaer Polytechnic Institute

Troy, NY

- Analyzed and visualized microbial locality datasets spanning 1,000+ enzymes, supporting biochemical and geobiological research.
- Curated and standardized 10+ archival scientific datasets for the NASA-funded Earth First Origins project, improving data accessibility for downstream analysis.
- Developed a diabetes ontology as part of a graduate team project, applying knowledge representation principles to biomedical domain modeling.

RESEARCH PROJECTS

POLO: Prebiotic Origins of Life Ontology | OWL, RML, SPARQL, Protégé, GraphDB

- First formal ontology for experimental prebiotic chemistry. OWL-based, reasoner-validated (HermiT/Pellet), FAIR-aligned, SPARQL-queryable. Encodes 184 experiments as 12,218 RDF triples via RMLMapper. Published with persistent PURL namespace (purl.org/polo). CC BY-NC 4.0.

EORE: Executable Online Resource of Experimental Data | Express.js, MongoDB, Python

- Data ingestion and management framework for prebiotic chemistry and origins-of-life experimental data. Captures conditions, chemical species, and provenance for 190+ experiments. Designed as a community-curated resource aligned with FAIR principles and NASA data standards.

Model LineUpper | Python, SHAP, scikit-learn

ACM IUI 2021

- Explainable AI system for multi-level ML model comparison. Evaluated in controlled study with professional data scientists. Supports model selection via SHAP-based feature importance and instance-level error diagnosis. 42 citations.

Speech Emotion Recognition Benchmarking | Python, FastAPI, PyTorch

- Benchmarking pipeline for SER models against IEMOCAP and MSP-Podcast. Identified dataset-model co-development bias in EmoNet-Voice Bench. Arousal accuracy ~37.8%, valence accuracy ~39.2% on IEMOCAP.

PUBLICATIONS

Peer-Reviewed / Submitted

- Narkar, S.U., Riggi, V.S., Rogers, K.L., Hendler, J.A. **POLO: A Formal Ontology for Experimental Prebiotic Chemistry**. *Submitted to ISWC, Resource Track*, 2026.
- Narkar, S.U., et al. **Model LineUpper: Supporting Interactive Model Comparison at Multiple Levels for AutoML**. *Proc. ACM IUI*, 2021.
- Boujibar, A., et al. (incl. Narkar, S.U.) **Cluster Analysis of Presolar Silicon Carbide Grains**. *Astrophysical Journal Letters*, 2021.

In Preparation

- Narkar, S.U., Riggi, V.S., Hendler, J.A., Rogers, K.L. **A Data Ingestion Framework for Experimental Prebiotic Chemistry**.
- Narkar, S.U., Riggi, V.S., Hendler, J.A., Rogers, K.L. **Predictive Modeling and Analysis of Prebiotic Chemistry Experimental Data**.

PRESENTATIONS

- **Invited Talk:** Leveraging Computational and Semantic Technologies in Understanding Prebiotic Chemistry. *GSA Annual Meeting*, 2023.
- **Talk:** Enabling Semantic Technologies and eScience for Exploring the Prebiotic Chemistry Experimental Landscape. *AbSciCon*, 2024.
- **Poster:** POLO and EORE: Ontology and Data Infrastructure for Prebiotic Chemistry. *AbSciCon*, 2026.

TEACHING & MENTORING

Teaching Assistant , Computer Science I, RPI	2023
Teaching Assistant , Econometrics in Big Data with R, RPI	2018
Undergraduate Research Advisor , RPI Undergraduate Research Program (3 students)	2025 – Present
Undergraduate Research Advisor , Rensselaer Research Experience Program	2019
Graduate Student Mentor (2 students)	2025 – Present
High School Research Mentor (1 student)	2025 – Present

HONORS & AWARDS

RPI Founders Award of Excellence, Rensselaer Polytechnic Institute	2018
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SERVICE & AFFILIATIONS

Interim Secretary-Treasurer , Geoinformatics & Data Science Division, GSA	2024
Student Officer , Geoinformatics & Data Science Division, GSA	2019 – 2023
Member , Geological Society of America; American Geophysical Union	